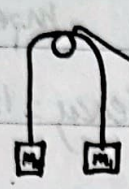
HW #10

1)



Given: $m_1 = 5.00\,kg$, $m_2 = 2.00\,kg$,

$\Delta y = 40\,cm = .4\,m$,

$\cancel{\Delta K_{mech}} = 0 = \Delta E_{mech}$

$\cancel{\Delta K_{mech} = K_f - K_i = 4}$

$\cancel{\Delta E_{mech} = \Delta K}$

1)

$-y_0 = -.4m$

$-y = 0$

Given: $m_1 = 5.00\,kg$, $m_2 = 2.00\,kg$,

$\Delta y = 40\,cm = .4\,m$,

$\Delta E_{mech} = 0$, $K = \frac{1}{2}mv^2$, $U = mgy$

$$\Delta E_{mech} = \Delta K + \Delta U$$

$$= \Delta K_1 + \Delta K_2 + \Delta U_1 + \Delta U_2$$

$$= (K_{1f} - K_{1i}) + (K_{2f} - K_{2i}) + (U_{1f} - U_{1i}) + (U_{2f} - U_{2i})$$

$$= \cancel{\tfrac{1}{2}v^2(m_1 - 0)}\left(\tfrac{1}{2}m_1 v_f^2 - \tfrac{1}{2}m_1 \cdot 0\right) + \left(\tfrac{1}{2}m_2 v_f^2 - \tfrac{1}{2}m_2 \cdot 0\right)$$

$$+ (0 - m_1 g y_0) + (m_2 g - y_0 - 0)$$

$$\Delta E_{mech} = \tfrac{1}{2}v_f^2(m_1 + m_2) + -g y_0 (m_1 - m_2)$$

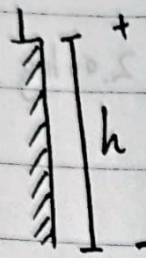
$$0 = \tfrac{1}{2}v_f^2(5+2)\,kg - 9.81\,m/s^2 \cdot .4\,m\,(\cancel{3-5}\ 5-2)\,kg$$

$$11.772 = \tfrac{7}{2}v_f^2\,kg$$

$$\tfrac{2}{7} \times 11.772 = v_f^2$$

$$v_f = \cancel{\sqrt{3.145}\ m/s}\ \sqrt{3.36}\ m/s$$

$$= \cancel{5.608\ m/s}\ 1.83\ m/s$$

2) 

Given: $W_{meta} = 3.76 \times 10^4 \, J/g$, $m_{person} = 70.0 \, kg$

$h = \Delta y = 433 m$, efficiency $= 15\%$

$\Delta U = mgy$ (mechanical)

$$\Delta U_{chem} = \frac{\Delta U_{mech}}{\text{efficiency}}$$

$$m_{weightlost} = \frac{\Delta U_{chem}}{W_{meta}}$$

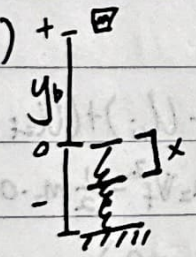
$$= \frac{mgy}{\text{efficiency}} \cdot \frac{1}{W_{meta}}$$

$$= \frac{70 kg \cdot -9.81 m/s^2 \cdot 433 m}{.15} \cdot \frac{1}{3.76 \times 10^4 J/g}$$

$$= -\frac{297341.1}{.15} J \cdot \frac{1}{37600 J/g}$$

$$= -1982274 J \cdot \frac{1}{37600 J/g}$$

$$m(g) = -52.72$$

3) 

Given: $m = 500 g = .5 kg$, $y_b = 60 cm = .60 m$,

$k_s = 120 N/m$

$U_b = mgy$

$\therefore$ U_b will transfer to the spring starting from $y = 0$

$\therefore$ ~~tt~~ $\Delta U_s = -U_b$

$\Delta U_s = -mgy$ $\quad \Delta U_s = \frac{1}{2} k x^2$

$$x = \sqrt{\frac{2U_s}{k}}$$

$$= \sqrt{\frac{-2mgy}{k}}$$

$$= \sqrt{\frac{-2 \cdot .5 kg \cdot -9.81 m/s^2 \cdot .60 m}{120 kg \frac{m}{s^2}/m}}$$

$$= \sqrt{4.905 \times 10^{-2} m^2}$$

$$= .22 m$$

4)

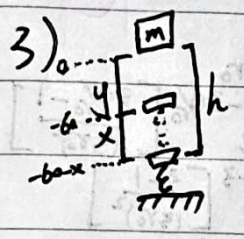
a) $r=1$ & $r=3$

b) $[r=1 \ \& \ r=4]$, $[r=4.5 \ \& \ r=5.5]$

c) $E_{mech} = U + K$

$$K = E_{mech} - U$$
$$= -4 - -6$$
$$= 2$$

3)



$$E_{mech} = 0$$
$$E_{mech} = \Delta U_b + \Delta U_s$$
$$= [U_{bf} - U_{bi}] + [U_{sf} - U_{si}]$$
$$0 = mg[-60-x-0] + \tfrac{1}{2}k[x^2 - 60 - x - -60]^2$$
$$= mg[-60-x] + \tfrac{1}{2}kx^2$$
$$-\tfrac{1}{2}kx^2 = mg[-60-x]$$
$$x^2 = -2mg[-60-x]/k$$
$$\cancel{x = \sqrt{-2mg[-60-x]/k}}$$
$$x^2 = [+2mg \cdot 60 + 2xmg]/k$$
$$0 = 120\frac{mg}{k} + 2x\frac{mg}{k} - x^2$$
$$x = .26\,m$$

5)

$$U(x) = -\int F(x)\,dx$$
$$= -\int_{\infty}^{x'} \frac{b}{x^2}\,dx$$
$$= -b\int_{\infty}^{x'} x^{-2}\,dx$$
$$= +bx^{-1}\Big|_{\infty}^{x'}$$
$$= +\frac{b}{x}\Big|_{\infty}^{x'}$$
$$= \frac{b}{x'} - \frac{b}{\infty}$$
$$= \frac{b}{x'}$$

6) Given: r_0 & U_0 are constant

$$F(x) = -\frac{d}{dx} U(x)$$

a) $$F(r) = -\frac{d}{dr} U(r)$$

$$= -\frac{d}{dr} -\frac{r_0}{r} U_0 e^{-r/r_0}$$

$$= +r_0 U_0 \frac{d}{dr} \frac{1}{r} \cdot e^{-r/r_0}$$

$$= +r_0 U_0 \left[\frac{-\frac{1}{r_0} e^{-r/r_0} \cdot r - e^{-r/r_0}}{r^2} \right]$$

$$= -r_0 U_0 e^{-r/r_0} \left[\frac{r/r_0 + 1}{r^2} \right]$$

b) while $r = r_0$

$$F(r) = -r_0 U_0 e^{-r_0/r_0} \left[\frac{r_0/r_0 + 1}{r_0^2} \right]$$

$$= -r_0 U_0 e^{-1} \left[\frac{1+1}{r_0^2} \right]$$

$$= r_0 U_0 e^{-1} \left[\frac{0}{r_0^3} \right]$$

$$= r_0 U_0 e^{-1} \cdot 0$$

$$= 0$$

X

$$= -r_0 U_0 e^{-1} \left[\frac{2}{r_0^2} \right]$$

$$= -2452.529$$

while $r = 3r_0$

$$F(r) = -r_0 U_0 e^{-3r_0/r_0} \left[\frac{3r_0/r_0 - 1}{(3r_0)^2} \right]$$

$$= r_0 U_0 e^{-3} \left[\frac{3-1}{(3r_0)^2} \right]$$

$$= r_0 U_0 e^{-3} \left[\frac{2}{9r_0^2} \right]$$

$$= 7.5 \times 10^{-27} \cdot e^{-3} \left[\frac{2}{2.025 \times 10^{-29}} \right]$$

$$= 3.73 \times 10^{-28} \cdot \frac{2}{2.025 \times 10^{-29}}$$

$$= \frac{2}{7.56 \times 10^{-57}}$$

X

$$F(r) = -r_0 U_0 e^{-3r_0/r_0} \left[\frac{3r_0/r_0 + 1}{(3r_0)^2} \right]$$

$$= -73.75$$

7)

$$E_{mech} = 0$$
$$= \Delta U + \Delta K$$
$$= U_f - U_i + K_f - K_i$$
$$= mg[2(y-h) - 2] + \frac{1}{2}mv^2 - 0$$
$$= g[2h + 2y - 2] + \frac{1}{2}v^2$$
$$0 = -2hg + 2yg - 2g + \frac{1}{2}v^2$$
$$2hg = 2yg - 2g + \frac{1}{2}v^2$$
$$=$$

$$E_{mech} = 0$$
$$= \Delta U + \Delta K$$
$$= U_f - U_i + K_f - K_i$$
$$= mg(2(a-h) - a) + \frac{1}{2}mv^2 - 0$$
$$= g(2a - 2h - a) + \frac{1}{2}v^2$$
$$0 = ga - 2gh + \frac{1}{2}v^2$$
$$2gh = ga + \frac{1}{2}v^2$$
$$-v^2 = (ga - 2gh)2$$
$$g(a-h) \leq -2ga + 4gh$$
$$ga - gh \leq -2ga + 4gh$$
$$a - h \leq -2a + 4h$$
$$3a \leq 5h$$
$$\cancel{h \leq a/3}$$
$$\frac{3a}{5} \leq h$$

$$F = \frac{mv^2}{r}$$

$\because$ mass swing around the peg

$$\therefore \frac{mv^2}{r} \geq mg$$
$$v^2 \geq gr$$
$$\because r = a - h$$
$$\therefore v^2 \geq g(a-h)$$

8)

$E_{mech} = 0$

$= U_f - U_i + K_f - K_i$

$= 0 - mgh + \frac{1}{2}mv^2 - 0$

$\frac{1}{2}mv^2 = mgh$

$v^2 = 2hg$

$v = \sqrt{2hg}$

$H = \frac{1}{2}at^2$

$t^2 = \frac{2H}{a}$

$t = \sqrt{\frac{2H}{a}}$

$D = vt$

$= \sqrt{2hg}\sqrt{\frac{2H}{a}}$

$\because a = g$

$= \sqrt{2hg} \cdot \sqrt{\frac{2H}{g}}$